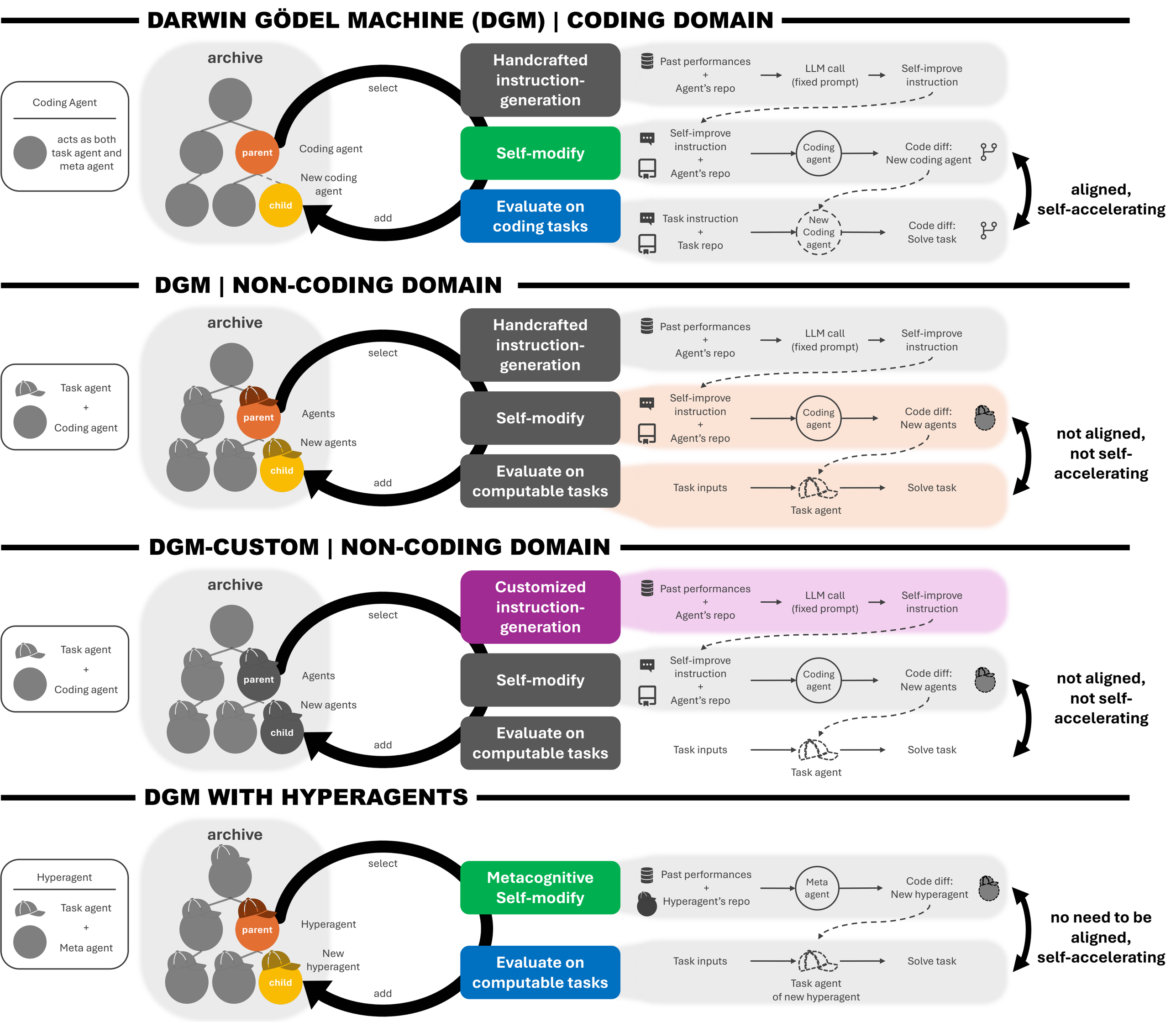
DARWIN GÖDEL MACHINE (DGM) | CODING DOMAIN
Coding Agent
acts as both task agent and meta agent
archive
parent
child
select
add
Coding agent
New coding agent
Handcrafted instruction-generation
Self-modify
Evaluate on coding tasks
Past performances + Agent's repo
LLM call (fixed prompt)
Self-improve instruction
Self-improve instruction + Agent's repo
Coding agent
Code diff: New coding agent
Task instruction + Task repo
New Coding agent
Code diff: Solve task
aligned, self-accelerating
DGM | NON-CODING DOMAIN
Task agent + Coding agent
archive
parent
child
select
add
Agents
New agents
Handcrafted instruction-generation
Self-modify
Evaluate on computable tasks
Past performances + Agent's repo
LLM call (fixed prompt)
Self-improve instruction
Self-improve instruction + Agent's repo
Coding agent
Code diff: New agents
Task inputs
Task agent
Solve task
not aligned, not self-accelerating
DGM-CUSTOM | NON-CODING DOMAIN
Task agent + Coding agent
archive
parent
child
select
add
Agents
New agents
Customized instruction-generation
Self-modify
Evaluate on computable tasks
Past performances + Agent's repo
LLM call (fixed prompt)
Self-improve instruction
Self-improve instruction + Agent's repo
Coding agent
Code diff: New agents
Task inputs
Task agent
Solve task
not aligned, not self-accelerating
DGM WITH HYPERAGENTS
Hyperagent
Task agent + Meta agent
archive
parent
child
select
add
Hyperagent
New hyperagent
Metacognitive Self-modify
Evaluate on computable tasks
Past performances + Hyperagent's repo
Meta agent
Code diff: New hyperagent
Task inputs
Task agent of new hyperagent
Solve task
no need to be aligned, self-accelerating